ECE 302 Formula Sheet

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Bipolar Junction Transistor (BJT)

$$i_C = I_s e^{v_{BE}/V_T} = \alpha i_E = \beta i_B \qquad i_C = I_s e^{v_{BE}/V_T}\left(1 + \frac{v_{CE}}{V_A}\right)$$

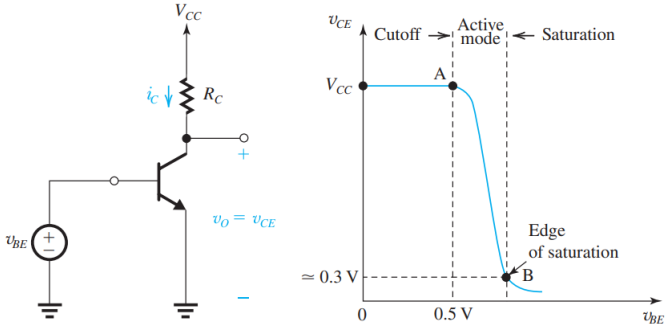
$$i_E = (\beta + 1)\, i_B = i_B/(1-\alpha) \qquad \alpha = \frac{\beta}{\beta+1} \qquad \beta = \frac{\alpha}{1-\alpha}$$

Forward Bias Conditions (npn):

$$v_{BE} = +0.7[V] \quad v_{CB} \geq -0.4[V] \implies v_{CE} \geq +0.3[V]$$

BJT Amplifiers

Skeleton Amplifier:



Skeleton Circuit Characteristics:

$$v_{CE} = V_{CC} - i_C R_C = V_{CC} - R_C I_S e^{v_{BE}/V_T}$$

$$A_v = \left.\frac{dv_{CE}}{dv_{BE}}\right|_{v_{BE}=V_{BE}} = \frac{-I_C R_C}{V_T} = -\frac{(V_{CC} - V_{CE})}{V_T}$$

Small Signal Parameters:

$$g_m = \frac{I_C}{V_T} = \frac{i_c}{v_{be}} \qquad r_\pi = \frac{v_{be}}{i_b} = \frac{\beta}{g_m} \qquad r_e = \frac{v_{be}}{i_e} = \frac{\alpha}{g_m} \approx \frac{1}{g_m}$$

Need $v_{be} << V_T$ for small signal approximation to be valid ($v_{be} \approx 10[mV]$ or less).

MOSFET Transistors

Device Parameters:

$$v_{OV} \equiv v_{GS} - V_{th} \qquad C_{ox} = \frac{\epsilon_{ox}}{t_{ox}},\ \epsilon_{ox} = 3\epsilon_0 \qquad |Q| = C_{ox}(WL)v_{OV}$$

$$k_n' = \mu_n C_{ox} \qquad k_n = k_n'(W/L) \qquad g_{DS} = \frac{1}{r_{DS}} = k_n v_{OV}$$

In triode or saturation $\implies V_{GS} > V_{tn}$. Otherwise, cutoff.

Saturation $\implies V_{GD} \leq V_{tn}$ or $v_{DS} \geq v_{OV}$.

Triode $\implies V_{GD} > V_{tn}$ or $v_{DS} < v_{OV}$

Saturation Region:

$$i_D = \frac{1}{2}k_n' \frac{W}{L} v_{OV}^2 \qquad i_G \approx 0,\ i_D \approx i_S$$

Channel Length Modulation:

$$i_D = \frac{1}{2}k_n' \frac{W}{L} v_{OV}^2 = \left(1 + \frac{v_{DS}}{V_A}\right) \text{ with } \lambda = \frac{1}{V_A}$$

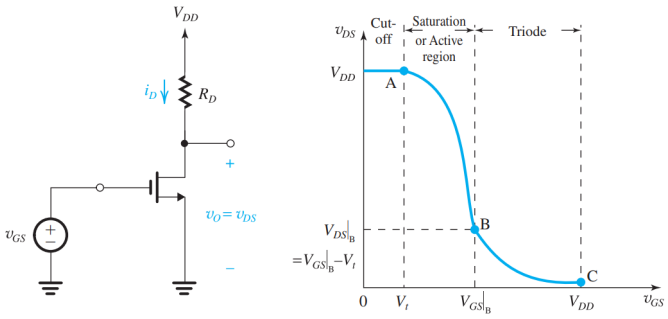
$$r_o = \frac{V_A}{I_D'} \text{ (i.e. I' without Early Effect)}$$

Triode Region:

$$i_D = k_n' \frac{W}{L}\left(V_{OV} v_{DS} - \frac{1}{2}v_{DS}^2\right)$$

MOSFET Amplifiers

Skeleton Amplifier:



Skeleton Circuit Characteristics:

$$v_{DS} = V_{DD} - i_D R_D = V_{DD} - \frac{1}{2}k_n R_D (v_{GS} - V_T)^2$$

$$V_{GS}|_B = V_t + \frac{\sqrt{2k_n R_D V_{DD} + 1} - 1}{k_n R_D} = V_{OV}|_B + V_T$$

$$A_v = \left.\frac{dv_{DS}}{dv_{GS}}\right|_{v_{GS}=V_{GS}} = -k_n\left(V_{GS} - V_t\right) R_D = -k_n V_{OV} R_D$$

$$\implies A_v = -\frac{I_D R_D}{V_{OV}/2} = -\frac{V_{DD} - V_{DS}}{V_{OV}/2}$$

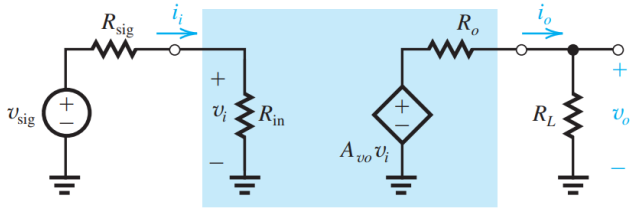
Small Signal Parameters:

$$g_m = \frac{i_d}{v_{gs}} = k_n\left(V_{GS} - V_t\right) = k_n V_{OV} = \frac{2I_D}{V_{OV}}$$

$$r_o = \frac{|V_A|}{I_D} \qquad i_d \approx i_s \qquad i_g \approx 0$$

Need $v_{gs} << 2V_{OV}$ for small signal approx. to be valid

General Voltage Amplifier



Voltage Amplifier Characteristics:

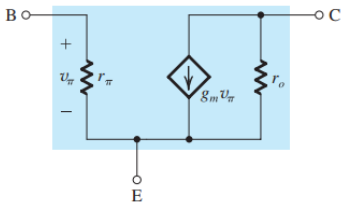
$$A_{v0} = \left.\frac{v_o}{v_i}\right|_{R_L=\infty} \qquad G_v = \frac{v_o}{v_{sig}} = \frac{R_{in}}{R_{in}+R_{sig}} A_{v0} \frac{R_L}{R_L+R_o}$$

$$A_v = \frac{v_o}{v_i} = A_{v0}\frac{R_L}{R_L+R_o} \qquad R_{in} = \frac{v_i}{i_i} \qquad R_o = \left.\frac{v_o}{i_o}\right|_{v_i=0\ R_L=\infty}$$

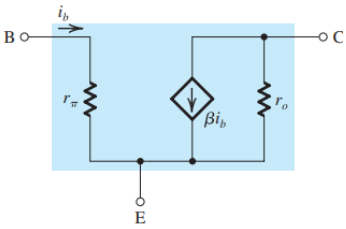
Small Signal BJT Models

Hybrid-π Model

■ $(g_m v_\pi)$ Version

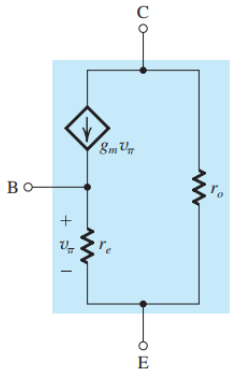


■ (βi_b) Version

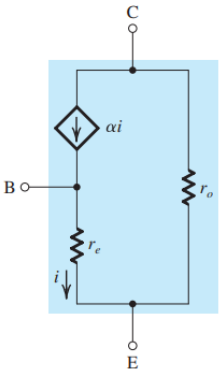


T Model

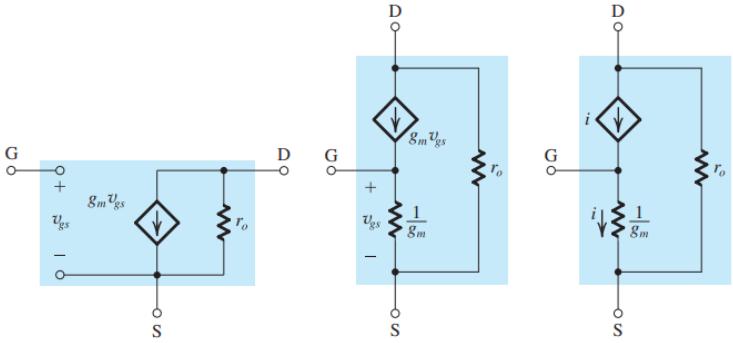
■ $(g_m v_\pi)$ Version



■ (αi) Version

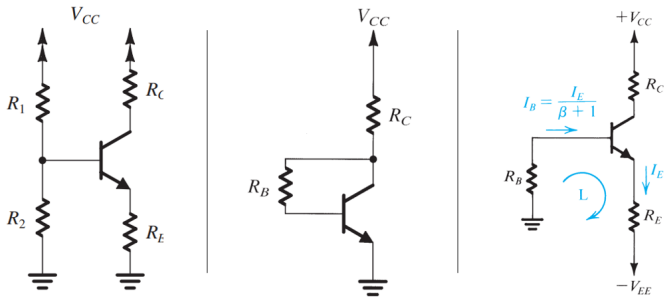


Small Signal MOSFET Models



Bias Circuits

Single Rail, Feedback, Dual Rail



Single Rail (Left)

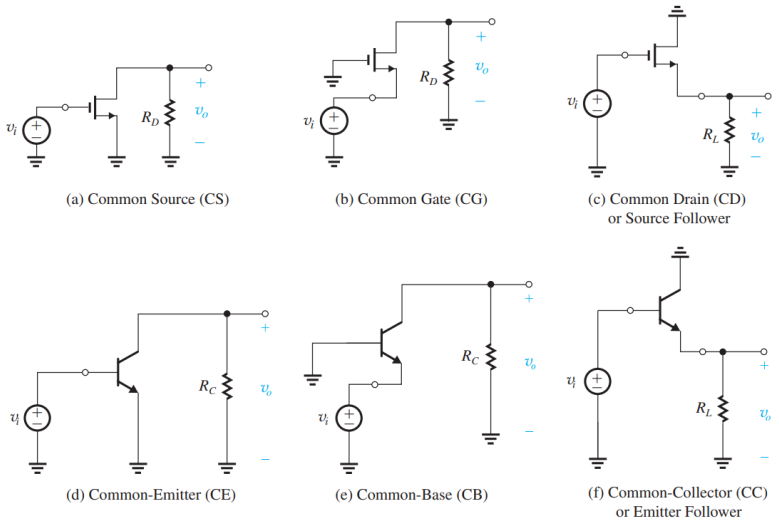
$$I_E = \frac{V_{BB}-V_{BE}}{R_E + R_B/(\beta+1)} \qquad V_{BB} = \frac{R_2}{R_1+R_2}V_{CC}$$

Feedback (Center)

$$I_E = \frac{V_{CC}-V_{BE}}{R_C + R_B/(\beta+1)}$$

Dual Rail (Right)

$$I_E = \frac{V_{EE}-V_{BE}}{R_E + R_B/(\beta+1)}$$



(a) Common Source (CS) (b) Common Gate (CG) (c) Common Drain (CD) or Source Follower

(d) Common-Emitter (CE) (e) Common-Base (CB) (f) Common-Collector (CC) or Emitter Follower

Table 7.5 Characteristics of BJT Amplifiers[a,b]

	R_{in}	A_{vo}	R_o	A_v	G_v
Common emitter (Fig. 7.36)	$(\beta+1)r_e$	$-g_mR_C$	R_C	$-g_m(R_C \parallel R_L)$ $-\alpha\dfrac{R_C \parallel R_L}{r_e}$	$-\beta\dfrac{R_C \parallel R_L}{R_{sig}+(\beta+1)r_e}$
Common emitter with R_e (Fig. 7.38)	$(\beta+1)(r_e+R_e)$	$-\dfrac{g_mR_C}{1+g_mR_e}$	R_C	$\dfrac{-g_m(R_C \parallel R_L)}{1+g_mR_e}$ $-\alpha\dfrac{R_C \parallel R_L}{r_e+R_e}$	$-\beta\dfrac{R_C \parallel R_L}{R_{sig}+(\beta+1)(r_e+R_e)}$
Common base (Fig. 7.40)	r_e	g_mR_C	R_C	$g_m(R_C \parallel R_L)$ $\alpha\dfrac{R_C \parallel R_L}{r_e}$	$\alpha\dfrac{R_C \parallel R_L}{R_{sig}+r_e}$
Emitter follower (Fig. 7.43)	$(\beta+1)(r_e+R_L)$	1	r_e	$\dfrac{R_L}{R_L+r_e}$	$\dfrac{R_L}{R_L+r_e+R_{sig}/(\beta+1)}$ $G_{vo}=1$ $R_{out}=r_e+\dfrac{R_{sig}}{\beta+1}$

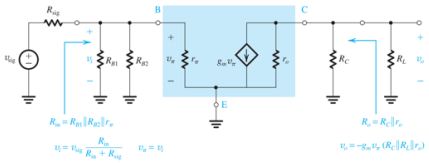
[a] For the interpretation of R_m, A_{vo}, and R_o refer to Fig. 7.34.
[b] Setting $\beta=\infty$ ($\alpha=1$) and replacing r_e with $1/g_m$, R_C with R_D, and R_e with R_s results in the corresponding formulas for MOSFET amplifiers (Table 7.4).

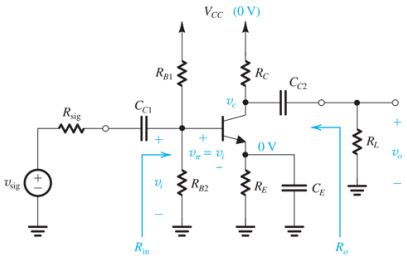
Table 7.4 Characteristics of MOSFET Amplifiers

	Characteristics[a]				
Amplifier type	R_{in}	A_{vo}	R_o	A_v	G_v
Common source (Fig. 7.35)	∞	$-g_mR_D$	R_D	$-g_m(R_D \parallel R_L)$	$-g_m(R_D \parallel R_L)$
Common source with R_s (Fig. 7.37)	∞	$-\dfrac{g_mR_D}{1+g_mR_s}$	R_D	$\dfrac{-g_m(R_D \parallel R_L)}{1+g_mR_s}$ $-\dfrac{R_D \parallel R_L}{1/g_m+R_s}$	$-\dfrac{g_m(R_D \parallel R_L)}{1+g_mR_s}$ $-\dfrac{R_D \parallel R_L}{1/g_m+R_s}$

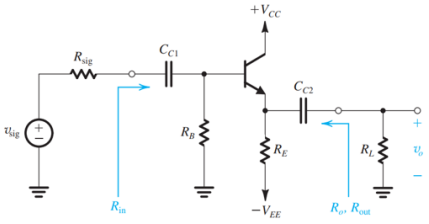
Basic Discrete Circuit BJT Amplifiers

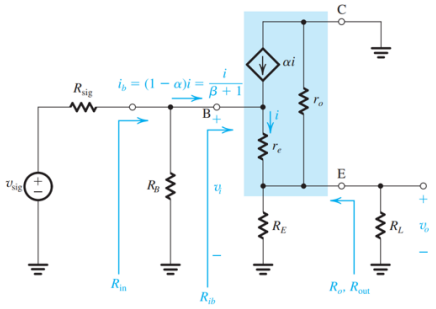
Common Emitter





Common Collector (Emitter Follower)





Common Base

